ITSM Lab: Comprehensive Post-Incident Review, 5-Whys Analysis & Blameless SRE Remediation

Duration : 90 Min.

Scenario

Following a severe 3-hour core banking ledger outage, the organization requires an exhaustive, audit-grade Post-Incident Review (PIR). Conduct an end-to-end retrospective: reconstruct the complete chronological timeline from telemetry and bridge notes, execute a multi-branch 5-Whys root cause analysis, identify technical and organizational contributing factors, and author a formal LTM PIR report with SMART corrective and preventive actions.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Reconstruct a minute-by-minute timeline across detection, mobilization, triage, fix, and verification. - Execute deep 5-Whys root cause analysis uncovering underlying systemic flaws. - Evaluate contributing factors across monitoring visibility, testing gaps, and runbook accuracy. - Formulate SMART preventive actions with assigned engineering owners and target delivery dates. - Calculate SLA impact, downtime cost, and error budget depletion. - Prepare executive summary for C-level leadership. - Update Knowledge Base and Known Error Database with lessons learned.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. Construction of exhaustive, minute-by-minute timeline integrating telemetry and bridge records: 3 marks
2. Execution of multi-branch 5-Whys analysis drilling to systemic architectural and process root causes: 3 marks
3. Identification of latent contributing factors across observability, deployment, and testing gates: 3 marks

4. Formulation of SMART corrective and preventive actions with assigned owners and realistic deadlines: 3 marks
5. Accurate quantification of business impact, MTTA/MTTR metrics, and error budget depletion: 3 marks
6. Authoring of high-impact executive summary tailored for non-technical enterprise stakeholders: 3 marks
7. Generation of runbook updates and Known Error Database entries to prevent recurrence: 2 marks

Total Score: 20 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.